

The boiler design of remote monitoring system based on the SCADA

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Abstract—A remote monitoring system has been designed in this article by using SCADA. The whole application based on B/S structure has been implemented. Specifically, the monitor signals of boiler dry spot's worker are been collected first, and a real-time database is established on the basis of the collected information. Then, using Msajax and Silverlight techniques to deal with the former collected data, the remote real-time monitoring function can be realized. In summary, these approaches could extend the system function and optimize the system structure. More importantly, it obtains the real-time data of web monitor system, which can draw the real-time curve of the dynamic systems and the curve of history, etc. Furthermore, it could achieve real-time brushless description.

Keywords—SCADA; B/S structure; dynamic real-time curve; msajax and silverlight; remote real-time monitoring; real-time brushless description

I. INTRODUCTION

With the full development of industrial technology, the flexibility of the Scada system used in industry was proposed higher requirements, and software model has been developed from C/S mode gradually shift to B/S mode. As Web users can through the remote network displays the monitored data, the system is not limited in the designated monitoring machine monitoring, greatly convenient for the user's access and management. The remote monitoring and control system is divided into two kinds: one kind is no monitoring system, which collects data to the remote computer processing directly. This kind of situation general data transmission distance is far away; The second type is the combination of field monitoring and remote monitoring, i.e. each unit via LAN connection. And configuration software are famous at home and abroad have begun to support distributed monitoring system based on Web, however there are some deficiencies and defects. A traditional Web technology to refresh the whole page and page refreshes are prone to jitter, influence the effect of the whole page, in the local data with just the page refresh when it is difficult to accurately in a timely manner. Although now there are some solutions, but they are not very satisfactory. This system adopts the solution using the Opc plus real-time database and Msajax Silverlight technology and successfully solved the problem. The dynamic real-time curve and historical curve graphics real-

time brushless description have been implemented.

II. REMOTE MONITORING TECHNOLOGY

Ajax (Asynchronous JavaScript and XML) is a kind of open Interactive Web application technology, It USES of the technology including the HTML (HyperText Markup Language, HyperText Markup Language) and CSS (Cascading Style Sheets, Cascading Style Sheets (CSS) to represent information, Using the DOM (Document Object Model, Document Object Model (DOM) and JavaScript on the page displays information interaction and dynamic display, Using XML (Extensible Markup Language, Extensible Markup Language), XSLT (Extensible Stylesheet Language Transformations, Extensible Stylesheet Transformations Language) and the XMLHttpRequest object asynchronously and exchange the data server and modify the data, Use JavaScript to put all things together. Ajax is a combination of many of the old technology, but this combination provides powerful function. Maintaining the Integrity of the Specifications

Traditional Web applications, directly from the browser sends a request to a Web server, Web server returned data directly to the Web browser updates page; Ajax applications by the Ajax engine through the XMLHttpRequest object sends a request, a Web server returned data, not directly to the browser, but rather to the Ajax engine, By the Ajax engine to update the page, the key is to Ajax's asynchronous data technology .

Traditional Web applications, send the request is synchronously, that is to say, after sending a request to the request return, it was a time when the user in addition to wait, nothing can be done, Therefore, early someone called WWW the World Wide Wait. While sent an Ajax request, can be asynchronously, namely sends the request in the background receives the request, Before the request return, users can still do the scrolling page, fill out the form, When the requested data returned, Ajax flicker-free to refresh the page to update part.

It's brought a good user experience, let the user feel and use the traditional application is not too big difference.

In addition, Ajax has a lot of UI (graphical user interface) controls are available, which is based on Ajax Web application is more and more like a traditional application, but also

traditional application incomparable superiority. Therefore, using Ajax technology to develop a telecontrol system based on Web, and will also be a remote system man-machine interface part of the mainstream development direction.

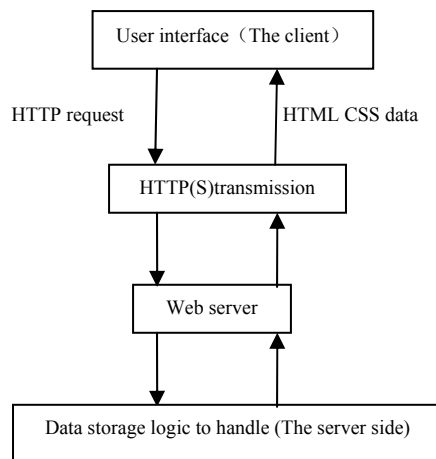


Fig. 1. Traditional Web applications

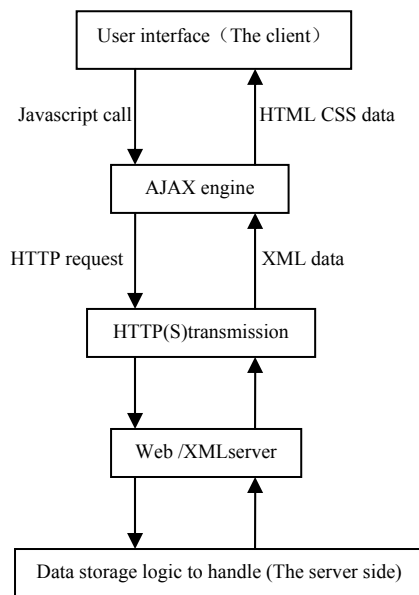


Fig. 2. AJAX applications

AJAX is not a technique, AJAX is just on behalf of the concept of Web programming a transformation, It is actually a combination of several kinds of technology. Each technology has its own way, together to create a powerful new technology. It consists of the following technology:

- the XHTML and CSS, to standardize performance;
- the DOM, is used to dynamically modify the content and structure of the document;
- XML for data exchange and processing;

- javascript, used for binding to the above several kinds of technology, makes it the work of the whole.

The advantages of AJAX technology is embodied in the following respects:

- It does not need to refresh the page. This is the greatest characteristic of AJAX, reduce user's psychology and the actual waiting time.
- It can reduce the financial burden on the server. Fetch the data "on demand" is the advantage of AJAX, can minimize redundant requests, reduce the burden of server.
- It can make full use of resources. Can be a lot of work on the server side is transferred to the client, easing the burden on the server, make full use of the bandwidth resources.
- It enables users to get a better experience. Users feel page submit without waiting, it felt like running locally.
- It is easy to maintenance maintenance. Separated from the application interface, easy to maintain.
- It is convenient for management. AJAX is based on standardized, and is widely supported by technology. The stage is based on the web, can any platform, do not need to download any plug-in or small programs.

The B/S structure of remote monitoring is made of the presentation layer, business logic layer and data layer. The presentation layer is realized through WWW browser by the user interface, the Web server service request is put forward by the browser, and the Web server requests are sent to the business logic layer. The business logic layer realizes the application program logic and completes the data processing, puts forward the data to the database server processes. The result of a database server to finish the request processing is sent to the Web server, then the Web server transmit the data to the client. The data layer is to provide services to the business logic layer, which can be for data storage and optimization, and to improve the efficiency of the system in this layer.

Remote monitoring is composed of the following stages:

- Data acquisition phase: Connect to the server via the Opc technology, Real-time data from the field into the real-time database.
- Data processing stages: For real time data in the database to the Web application on the server for processing.
- Dynamic display stage: Through Ajax technologies such as real-time display dynamic data on the client.
- User behavior to obtain phase: Complete control of the Web client object, get the user to control behavior.
- Control feedback phase: Completion of controlled object and parameters back to the Web server application, complete the processing.

- Control in the implementation stage: Complete application control of the field devices, realize the control of user intent.

AJAX application development basic processes: initialization object, send request to server receives, the server returned to the client receiving, modify the client page content. Ajax-based Web applications need to write a certain amount of javascript client script. When the workload is bigger, in order to alleviate the difficulty and complexity of the AJAX development, can consider to use the existing AJAX development framework, to improve the efficiency of development. AJAX development framework is divided into a browser-based application framework and application framework based on server two kinds. Browser-based application framework can be applied to any situation, you can use any server-side language (such as ASP, JSP, PHP, etc.) to dynamically generated page content. The application framework based on server often for a specific platform, for example, WebORB for Java is developing AJAX and Flash based client application development platform.

B/S (Browser/Server) structure namely the Browser and Server structure. Under this structure, The user interface is working on the WWW browser. Only few transaction logic in the front-end (Browser), but the main transaction logic on the Server side (Server), forming the so-called three layer structure. This greatly simplifies the client computer load, reduce the cost of system maintenance and upgrade and workload, To reduce the overall cost of users. User in the client work through WWW browser can achieve business.

Especially after the cross-platform language appeared, B/S structure management software is convenient, fast and high

Effect. adopts B/S structure of remote monitoring system to keep up with the pace of the rapid development of computer hardware and software level upgrade, Meet the needs of construction in different periods, different manufacturers to develop various systems interconnection between the application system software, information sharing and interoperability requirements, at the same time to adapt to the rapid development of Internet, improves performance of the system can be expanded, and can satisfy the requirement of the system in different platform to run. The system is a distributed power remote monitoring system of multipoint multilayer system principle as shown in figure 3, divided into the scene monitoring layer, bureau station monitoring layer, center monitoring, remote monitoring layer. On-site monitoring layer draw station monitoring using C/S and B/S combination of, Make full use of the resources of the enterprise local area network (LAN), as much as possible to increase both the efficiency and easy maintenance and extension Malleable. The center monitoring and remote monitoring layer adopts B/S mode, and realize the cross-platform, distributed and remote monitoring system performance. The scene monitoring layer is made up of acquisition module, communication module and industrial control. Bureau of station monitoring layer is monitored by the Web server, database server, client, printers, switches, routers, radio and other parts. Center monitoring layer consists of a Web server, database server, disk array, monitoring client, printers, routers, etc. Remote monitoring

layer is made up of any one computer can connect to the Internet, By the user access layer at the centre of the Web server to realize the monitoring of equipment, the user can to monitor equipment online at home, So as to realize the unattended equipment or less unattended. The whole monitoring system adopted B/S model, achieved the distribution in different regions across the Internet network of distributed remote monitoring of equipment, And provides data analysis and forecasting warning function, convenient for users for equipment fault diagnosis, Achieve the network load, reduces the network failure, in addition to realize dynamically add or delete the equipment, to solve the problem of equipment add or delete users, So as to improve the monitoring efficiency and the production efficiency of the enterprise, to make the enterprise the purpose of safe and efficient production.

The client is based on Web way, which maintains operating mainly on the server side. It is not high requirements for hardware conditions. Therefore it is easy to adjust system and updated. A change will not affect the other layers. This system just uses the Web based remote monitoring mode.

III. DESIGN OF THE SYSTEM

System is mainly for the boiler parameters such as pressure, flow and temperature data real-time acquisition and remote monitoring. This system design a front-end equipment remote monitor based on Ethernet communication protocols. It can be connected to the Internet for remote monitoring of unattended independent equipment and have the equipment abnormal alarm function. Random provide software interface, can easily form a remote monitoring system, realize the equipment environment, power supply and working condition of security devices such as remote monitoring. The local host monitoring computer parts of this system based on Microchip company developed MPLAB software platform developed by using c # language. It makes full use of the c # programming features of high efficiency and good performance. And it is developed a set based on the Opc server data real-time acquisition and monitoring.

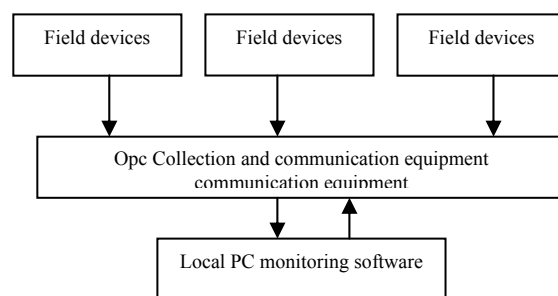


Fig. 3. Local PC system structure diagram

This system control aspects, can be manual or automatic control. When work environment parameters overrun field monitoring equipment to equipment failure occurs, the monitor output port through light lamp as a fault occurs. In manual control mode, the MCU upload alarm information to feedback to monitoring server users, prompt control accordingly. After PC receives the corresponding instruction manual control, it sends the data to the front-end equipment final control remote

monitor. In automatic mode, the MCU IO pin to pass direct control of the corresponding relay to control the output, and return up a machine control results. System is under a machine of S600 data collector for Siemens. S600 can complete data collection and the programmable function. It is through the Opc protocol communicates with local PC monitoring software, and implements the function of real-time database server through the network connection.

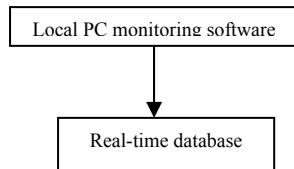


Fig. 4. Real-time communication structure

Application of real-time data achieves in complete read, data processing and data storage are processed simultaneously. After data processing, it will be a local database for subsequent report generation and graphic generation. When it gets the latest data from real-time database. Another processed directly through the Ajax technology to the client, the client data updated in real time.

Centralized monitoring to multiple terminals, the system design all terminals are directly connected to the Ethernet, can easily realize the remote centralized management. Since most single-chip computer doesn't have Ethernet interface, so the serial port is added in the front-end equipment remote monitor Ethernet conversion module, changes the traditional serial interface communication for Ethernet communications. Monitoring terminal information will be collected through MCU serial port sent to the conversion modules, to Ethernet server directly from this module. Front end equipment of the remote monitor serial port Ethernet conversion module in the network takes up an IP address and port number accordingly. When the malfunction point to point, the server based on Ethernet serial conversion module's IP address and port number automatic identification each monitory point, easy to operation and maintenance.

The client has three components:

- Real-time dynamic display module: Mainly through the Ajax technology and the Web server for local data exchange.
- Reports and historical curve module: Mainly through the local database to read according to the customers' demand to realize the data of the specified range of statements and the formation of history curve.
- Feedback control module: by adjusting and setting control parameters to the control requirements of users back to the Web server, the Web server software communicates with local PC monitoring software realize the control of user intent.

Traditional acquisition and monitoring system is that provide users with a powerful online control and better interface display, But affected by geographic, leave the production site, the user will know nothing about the

production status. The traditional B/S structure of the system is limited by server technology, real-time data due to repeated refreshes the page becomes scintillation and affect beautiful. To the data acquisition and monitoring system of production real-time data accurate and beautiful (page no refresh) show within the local area network (LAN), so using ASP.NET AJAX framework request server Web Service related method, adopts the OPC interface from the industrial field to process real time data in a relational database.

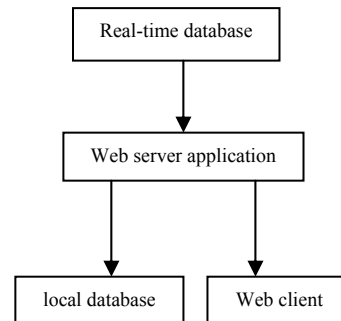


Fig. 5. Server data processing structure

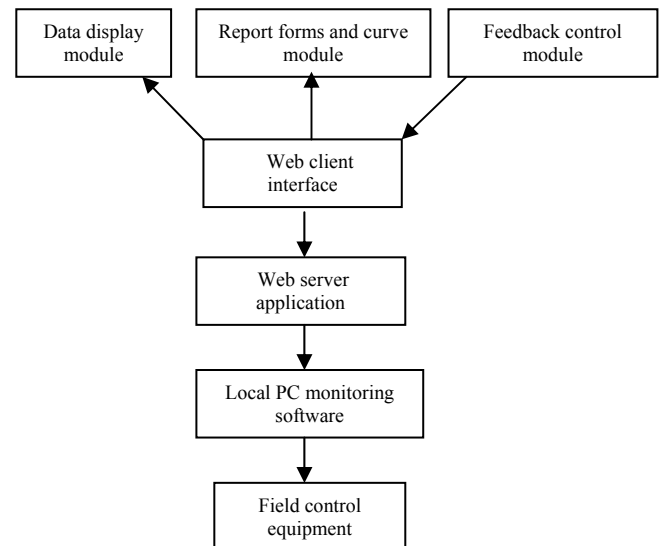


Fig. 6. Client function structure diagram

IV. ACHIEVEMENT

This article mainly to the front-end equipment remote monitor designed for on-site monitoring hardware, through the development of Web remote real-time monitoring system based on Ajax applications, real-time monitoring of the temperature and humidity. Centered on hardware reliability, field data collection and storage, temperature and humidity in excess of the alarmPassive control, provide test contact control content, and the result of the testThe corresponding analysis.

When system runs through in the control room is decorated in the region of the front-end equipment remote monitor and connect to the Internet, software applications running on the

server system, realize to the scene of the temperature and humidity data storage, monitoring and control.

Because of complicated field environment, there are many uncertain factors, so in this system all the front-end equipment after a long time in the current detection; To its high temperature resistance, anti-aging experiments, but also for the input, output, and data communication ports and so on a series of tests, so as to ensure the stability and reliability of the system hardware, can undertake the environmental monitoring requirements of the scene.

The PC monitoring software mainly uses c # development of Opc client read data, and adds the use for Opcw in the c # development environment. The library is as part of the .Net library, instead of belonging to the Com library. Although Siemens to provide for .Net platform support, these libraries are still difficult to programming. There is much number of data transmission in the managed and unmanaged area, therefore we develop a class library to simplify programming.

- Create a server interface
Server=News connection.synserver(Sconnection
.servertype.opc_Simatic Net);
- Open the connection
String err;
Server.open(out err);
- Add a group
Server.addgroup("gl1",1,350,out err);
- Add item
String[] m1={"S7:[one]MB1","S7:[one]MW3"};
Server.additems("gl1",m1,nt);
- Read and write data
Server.write("maiker",nt,obj);

The remote monitoring server uses the Windows server 2008 + IIS7 as operation platform, and uses c # + ajax + Silverlight technology to develop. Core classes are as follows:

- Public class real-time image maker ;
Complete real-time curve drawing.
- [Servcie contract][operation contract]&.ajax
settings.content type=" application/json"
Through to the JQuery calls brushless Wcf has realized
the data update.
- Xelement getpicture(xelement xml, string newpoint,
string attribute, string value) ;
Implements the graphic drawing.

Client browser side namely, monitoring and control system based on Web B/S structure of the client, without having to install any software Use the browser to browse the site equipment, environment of various kinds of real-time information, and store historical information, operation simple and convenient, flexible, intuitive and clear all kinds of information, has friendly man-machine interface

Remote monitoring system is a large amount of data, the longer, the more data. If you look at the data, every item of

data query to refresh the page, then, will see a data may need to wait a long time. But this system is through AJAX for database information, in the background when the user click to view one device monitoring information, will automatically find out the corresponding data values, show on the page, do not need to refresh, user experience is very good.

After a lot of testing and practical application, this system successfully on ASP.NET platform using AjaxReal-time display technology of Web based remote real-time monitoring and control system, B/S mode to realize the real-time monitoring of temperature and humidity on the spot, to effectively manage data, design button on a page by controlling the output port to control alarm, flicker-free page, and has faster response speed, strong real-time performance, has reached the requirement of real-time monitoring, and has a good man-machine interactive.

Often when we need to see a graphical reports, in order to make some decisions, We need the curve graphics. Curve graphics can put inside a month or a few months of data, by means of chart. Monitoring the page itself is large, the amount of data combined with several months of data processing on the server, using the traditional Web method is very awkward. This system solved the problem, through asynchronous reads data from a database to come over, don't need to refresh the page, you can display curve chart again.

V. CONCLUSION

A remote monitoring system has been designed in this article by using SCADA. The whole application based on B/S structure has been implemented. Specifically, the monitor signals of boiler dry spot's worker are been collected first, and a real-time database is established on the basis of the collected information. Then, using Msajax and Silverlight techniques to deal with the former collected data, the remote real-time monitoring function can be realized. In summary, these approaches could extend the system function and optimize the system structure. More importantly, it obtains the real-time data of web monitor system, which can draw the real-time curve of the dynamic systems and the curve of history, etc. Furthermore, it could achieve real-time brushless description. The designed system has run normally more than a year, and the three-dimensional graphics and real-time image transmission efficiency has a very good improvement in the practical applications.

By the client we implemented without a page refresh to update Web data of real-time data acquisition and monitoring system. Production managers no longer need to visit the production site to watch, only need one can connect in the Internet browser, without the need to install any third-party configuration software can real-time guide industrial production. The whole system can remote access, users are far away from the scene, can monitor the production of everywhere. In addition, a user can monitor more than one process at the same time, the process control personnel no longer overwhelmed; Multiple users can also monitor the same process, realize the data of openness and transparency of the process signal.

This system has very extensive application prospect in the remote monitoring system, is realized in a wide range of industrial information network and control network integration, to achieve the real meaning of remote monitoring and control to the factory, environment monitoring, water management, power grid monitoring play a huge role in promoting all aspects of development, the development of the "Internet of things" in the future also has a certain practical significance.

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